

Hasil Program

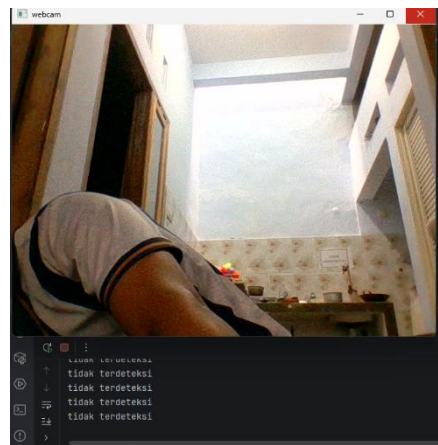
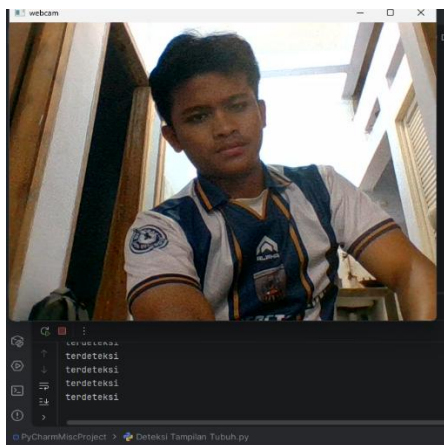
1. Percobaan A (Deteksi Anggota Tubuh)

Program

```
import cv2
import mediapipe as mp
mpose = mp.solutions.pose # inisiasi media pipe pose
pose = mpose.Pose()
cap = cv2.VideoCapture(0) # Video dari webcam

while True:
    success, img = cap.read() #pembacaan image
    imgrgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB) #Konvigurasi warna dari
    bgr ke rgb
    hasil = pose.process(imgrgb) #ekstraksi dari image
    if hasil.pose_landmarks:
        print ("terdeteksi")
    else:
        print ("tidak terdeteksi")
    cv2.imshow("webcam", img)
    cv2.waitKey(1)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break
cap.release() #tutup webcam dan jendela tampilan saat q ditekan
cv2.destroyAllWindows()
```

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2. Percobaan B (Pose Landmark)

Program

```
import cv2
import mediapipe as mp
mpose = mp.solutions.pose
pose = mpose.Pose()
mdraw = mp.solutions.drawing_utils
cap = cv2.VideoCapture(0)

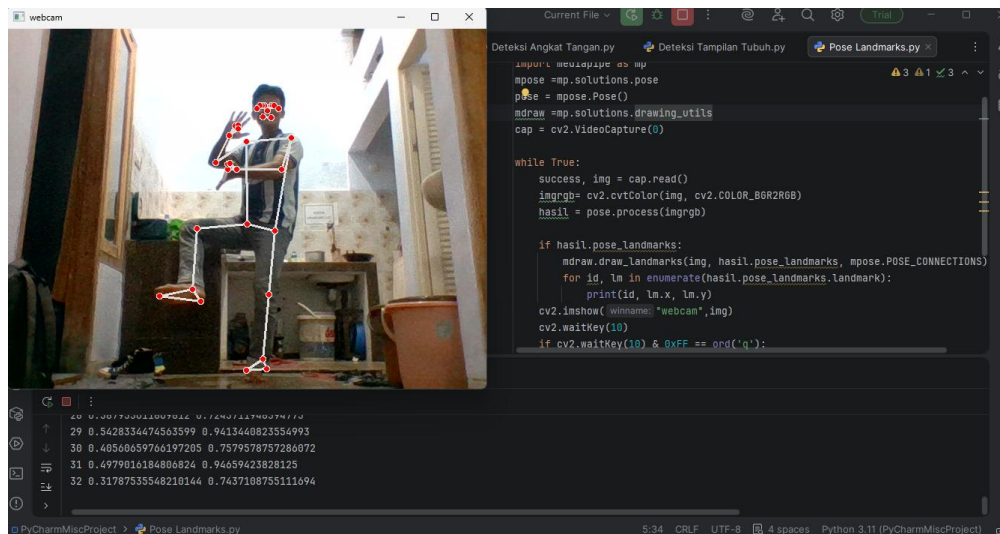
while True:
    success, img = cap.read()
    imgrgb= cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    hasil = pose.process(imgrgb)
```

```

        if hasil.pose_landmarks:
            mdraw.draw_landmarks(img, hasil.pose_landmarks,
mpose.POSE_CONNECTIONS)
            for id, lm in enumerate(hasil.pose_landmarks.landmark):
                print(id, lm.x, lm.y)
            cv2.imshow("webcam",img)
            cv2.waitKey(10)
            if cv2.waitKey(10) & 0xFF == ord('q'):
                break
        cap.release()
        cv2.destroyAllWindows()

```

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3. Latihan (Deteksi Gerakan Tangan)

Program

```

import cv2 #library open cv untuk akses kamera, pengolahan gambar, dan
tampilan video
import mediapipe as mp # library computer vision untuk deteksi pose, tangan,
wajah, dll
mpose = mp.solutions.pose # shortcut modul pose dari mediapipe
pose = mpose.Pose() # mengaktifkan model deteksi pose tubuh
mdraw = mp.solutions.drawing_utils # utility untuk menggambar titik landmark
tubuh
cap = cv2.VideoCapture(0) # mengaktifkan kamera pc

while True:
    success, img = cap.read()
    if not success:
        break
    imgrgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    hasil = pose.process(imgrgb)
    # Proses deteksi
    if hasil.pose_landmarks:
        mdraw.draw_landmarks(img, hasil.pose_landmarks,
mpose.POSE_CONNECTIONS) # Skleton Tubuh
        lm = hasil.pose_landmarks.landmark

        left_shoulder = lm[mpose.PoseLandmark.LEFT_SHOULDER]
        right_shoulder = lm[mpose.PoseLandmark.RIGHT_SHOULDER]

```

```

    left_wrist = lm[mpose.PoseLandmark.LEFT_WRIST]
    right_wrist = lm[mpose.PoseLandmark.RIGHT_WRIST]
# Deteksi tangan terangkat
    left_hand_up = left_wrist.y < left_shoulder.y
    right_hand_up = right_wrist.y < right_shoulder.y
    status = "Tidak Ada Tangan Terangkat"
    if left_hand_up and right_hand_up:
        status = "Dua Tangan Terangkat"
    elif left_hand_up:
        status = "Tangan Kiri Terangkat"
    elif right_hand_up:
        status = "Tangan Kanan Terangkat"
    cv2.putText(img, status, (50, 50), cv2.FONT_HERSHEY_SIMPLEX, 1, (0,
0, 0), 2)

    for id, lm in enumerate(hasil.pose_landmarks.landmark):
        print(id, lm.x, lm.y)
    cv2.imshow("webcam", img)
    if cv2.waitKey(10) & 0xFF == ord('q'):
        break
cap.release()
cv2.destroyAllWindows()

```

Hasil

